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Inventor(s)	Jones; Darren

Adjustable eyewear restraints, systems, and methods

Abstract

An adjustable eyewear restraint comprises a first strap having a first adjustment end and a first eyewear coupling end, and a second strap having a second adjustment end and a second eyewear coupling end. The straps are slidably and frictionally engageable to one another to allow adjustment of a length of the restraint by moving adjustment ends of the straps relative to each other. The straps can be slidably interwoven to each other about opening(s) of respective adjustment ends. The opening(s) slidably receive an opposing strap. A strap has a loop, coupleable to eyewear, that comprises first and second edges attached together to form an attachment portion substantially in-line with the first edge. A frictional element is frictionally and slidably coupleable to an earpiece to retain the loop and allow adjustment of a length of an eyewear retainer. Associated systems and methods are provided.

Inventors:	Jones; Darren (American Fork, UT)
Applicant:	Alpine Innovations LLC (Lehi, UT)
Family ID:	70280508
Assignee:	Alpine Innovations LLC (Lehi, UT)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4391498	12/1982	Rengstorff	351/111	G02C 5/2209
7467867	12/2007	Williams	D16/339	G02C 3/006
7845795	12/2009	Williams	2/452	G02C 3/003
D669115	12/2011	Kalbach	D16/339	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2854471	12/2003	FR	G02C 3/003

OTHER PUBLICATIONS

English translation of FR-2854471. (Year: 2004). cited by examiner

Primary Examiner: Lei; Jie

Attorney, Agent or Firm: Thorpe North & Western, LLP

Background/Summary

PRIORITY DATA (1) This application is a continuation of U.S. patent application Ser. No. 17/157,937, filed Jan. 25, 2021, which is a continuation of U.S. patent application Ser. No. 16/579,728, filed Sep. 23, 2019, which is a continuation of U.S. patent application Ser. No. 15/287,645, filed Oct. 6, 2016, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/237,970, filed on Oct. 6, 2015, each of which are incorporated herein by reference.

BACKGROUND

(1) Eyewear can have a tendency to become misaligned or maladjusted on a user, rendering the eyewear less effective or inoperable. Even worse, during rigorous physical activities, such as playing a sport, eyewear can be knocked or otherwise thrown from a user's face. Yet other environments, such a laboratories or workshops may require the frequent donning and removal of eyewear. In such circumstances as well as others, it may simply be convenient for a user to have a designated place close at hand to keep the eyewear for ready use.

(2) Accordingly, a variety of fastening mechanisms, straps, lanyards, and the like have been used to secure eyewear about a user, including in an operable and desirable position. Many of such fasteners are of a single fixed length and merely prevent the eyewear from falling to the ground by anchoring the eyewear to a user's neck. Where the fastener is adapted to secure the eyewear in an operable position (i.e. on a user's face), most often it is difficult or impossible to simply and easily move the eyewear to a non-operable position without completely disassociating the eyewear from the user. This is often due to the difficulty of the mechanism provided of tightening and loosening the fastener.

SUMMARY

(3) Accordingly, the present inventor has identified a need for eyewear fasteners that are capable of both securing the eyewear in an operable position and are simple and easy to adjust in order to allow the eyewear to be easily moved from the operable position to a non-operable position in some embodiments without effectively untethering the eyewear from the user. Adjustable eyewear restraints, and associated methods, are described herein that can both secure the eyewear and be easily adjustable to allow the eyewear to be moved and adjusted in length as desired by a wearer.

(4) In one example there is provided an adjustable eyewear restraint, comprising a first strap comprising a first adjustment end and a first eyewear-coupling end, with said first eyewear coupling end being coupleable to a first eyewear frame portion. A second strap comprises a second adjustment end and a second eyewear-coupling end, with said second eyewear coupling end being coupleable to a second eyewear frame portion. Thus, said first strap and said second strap being slidably engageable to one another to allow adjustment of a length of the restraint by moving at least one of the first adjustment end and the second adjustment end relative to the other one. In some examples, the first adjustment end of the first strap includes a first opening slidably frictionally engageable to the second strap, and the second adjustment end of the second strap includes a second opening and a third opening. Each of the second and third openings can be slidably frictionally engageable to the first strap. Each opening can be sized and shaped to provide sufficient frictional force, between the opening and the respective strap slidably engageable there through, to retain the straps to each other about a wearer's head. In some examples, each opening has a cross sectional area smaller than a cross section area of the respective strap slidably engageable through the respective opening. In some examples, each adjustment end comprises a pull tab positioned on the same side of the retainer when the straps are slidably engaged to each other. In some examples, the straps are slidably engaged to each other such that a length of the retainer is shortened when the adjustment ends are pulled away from each other, and such that a length of the retainer is lengthened when the adjustment ends are pulled toward each other. The straps can be slidably interwoven to each other about respective adjustment ends to form a substantially planar retainer configuration. In some examples, each opening comprises at least one of an aperture, a slot, or a loop.

(5) In one example, the first strap comprises a first male attachment and a first plurality of female attachments, and the second strap comprises a second male attachment and a second plurality of female attachments. The first plurality of female attachments is selectively interlockable to the second male attachment, and the second plurality of second female attachments is selectively interlockable to the first male attachment component.

(6) In one example, each strap comprises a first edge and an opposing second edge attached together, proximate respective eyewear coupling ends, to form a loop that receives a respective earpiece. The edges are attached to form an attachment portion substantially in-line with the first edge. In one example, the first edge comprises an upper edge extending along a length of the strap, and the second edge comprises a lower edge opposite the upper edge. In one example, the second edge has a length shorter than a length of the first edge. In one example, the attachment portion secures the first edge to the second edge by at least one of stitches, adhesive, fasteners, liquid plastic, and combinations thereof. In one example, the attachment portion comprises a predetermined length that is approximately a length of the second edge. In one example, the predetermined length is between approximately $\frac{1}{8}$ inch and $\frac{3}{4}$ inch. In one example, said loop is sized and shaped to frictionally receive and retain the earpiece. In one example, said loop is sized and shaped to loosely receive the earpiece, wherein an external frictional element is required to retain the earpiece within the loop.

(7) In one example there is provided an eyewear restraint system comprising a strap having an eyewear-coupling end slidably coupleable to an earpiece. The eyewear-coupling end can have a loop sized and shaped to allow a portion of the earpiece to pass through the loop. A frictional element is coupleable to the earpiece and positionable adjacent the loop of the strap, with said frictional element having a size and shape sufficient to frictionally engage the earpiece and retain the earpiece through the loop. In some examples, the frictional element has a perimeter body portion that defines a cross sectional area larger than a cross sectional area of an aperture of the loop. In some examples, the frictional element comprises an O-ring, or other device with an aperture, (e.g. a wedge shaped device) comprised of a compliant material and having an aperture sized and shaped to receive the earpiece such that the frictional element is frictionally and slidably coupleable to the earpiece for adjustment of the eyewear coupling end.

(8) In one example, the frictional element comprises an earpiece retainer having a first end and a second end. The first end is positioned through the loop and having an opening sized and shaped to removably attach the earpiece retainer to the earpiece, and the second end is positioned at least partially within the loop and having an enlarged interfacing portion that frictionally couples the earpiece retainer within the loop.

(9) In one example, the frictional element comprises a panel attached to the strap adjacent the loop, the friction panel having at least one opening configured to retain the earpiece.

(10) In one example there is provided an adjustable eyewear restraint system comprising a pair of eyewear having a first earpiece and a second earpiece. A first adjustable eyewear restraint comprises a pair of straps, each strap having a loop slidably coupled to a respective earpiece of the pair of eyewear. The pair of straps are slidably and frictionally engaged to one another to allow adjustment of a length of the first adjustable eyewear restraint by moving at least one of the straps relative to the other strap. A second adjustable eyewear restraint comprises a pair of frictional elements. Each frictional element is slidably and frictionally coupled to a respective earpiece to allow adjustment of a length of the first adjustable eyewear restraint by moving at least one of the frictional elements relative to the respective earpiece. In some examples, each strap comprises an inner planar surface facing the pair of eyewear and an outer planar surface positioned opposite the inner planar surface. Each strap comprises a pull tab positioned adjacent the outer planar surface of a respective straps such that the pull tabs are graspable proximate the outer planar surface of each strap.

(11) In some examples, the straps are comprised of a positively buoyant material having a selected volume and density sufficient to impart positive buoyancy on the pair of eyewear when the straps and eyewear are in fresh or salt water. The positively buoyant material can comprise a polymeric and/or fabric material (e.g. neoprene) material, and the pair of eyewear can weigh at least 30 grams.

(12) In one example there is provided a method of adjusting a length of an eyewear restraint. The method can comprise pulling a first adjustment end of a first strap away from a second adjustment

end of a second strap to shorten a length of the restraint and tighten an eyewear restraint about a wearer's head when coupled to a pair of eyewear. The first strap can be slidably and frictionally engaged to the second strap. The method can comprise pulling the first and second adjustment ends towards each other to lengthen a length of the restraint to loosening the restraint about the wearer's head. The method can comprise disposing a first earpiece of the pair of eyewear through a first loop of the first strap such that the first earpiece extends at least partially through the first loop. The method can comprise coupling a first frictional element to the first earpiece proximate the first loop to retain the first earpiece through the first loop. The first frictional element is slidably adjustable along the first earpiece to adjust a length of the eyewear restraint.

(13) The method can comprise disposing a second earpiece of the pair of eyewear through a second loop of the second strap such that the second earpiece extends at least partially through the second loop. The method can comprise coupling a second frictional element to the second earpiece proximate the second loop to retain the second earpiece through the second loop. The second frictional element is slidably adjustable along the second earpiece to adjust a length of the eyewear restraint.

(14) In one example there is provided a method of making an eyewear restraint. The method can comprise forming a first strap comprising a first adjustment end and a first eyewear-coupling end, with said first eyewear coupling end being coupleable to a first eyewear frame portion. The method can comprise forming a second strap comprising a second adjustment end and a second eyewear-coupling end, with said second eyewear coupling end being coupleable to a second eyewear frame portion. Thus, said first strap and said second strap are slidably engageable to one another to allow adjustment of a length of the restraint by moving at least one of the first adjustment end and the second adjustment end relative to the other one. The method can comprise forming a loop by securing a first strap edge to a second strap edge to form an attachment portion, wherein the attachment portion is substantially in-line with the first strap edge.

(15) There has thus been outlined, rather broadly, general invention features so that the detailed description thereof that follows may be better understood, and so that the present contribution to the art may be better appreciated. Other invention features will become clearer from the following detailed description, taken with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a perspective view of an adjustable eyewear restraint coupled to a pair of eyewear in accordance with one example;

(2) FIG. 1B is a perspective view of an adjustable eyewear restraint coupled to a pair of eyewear in accordance with one example;

(3) FIG. 2A shows various views of a strap of the restraint of FIG. 1B in accordance with one example;

(4) FIG. 2B shows various views of a strap of the restraint of FIG. 1B in accordance with one example;

(5) FIG. 3A is a perspective view of the adjustable eyewear restraint of FIG. 1B in accordance with one example;

(6) FIG. 3B illustrates adjustability of the adjustable eyewear restraint of FIG. 1B in accordance with one example;

(7) FIG. 4A shows inner and outer views of the adjustable eyewear restraint of FIG. 1B in an lengthened state in accordance with one example;

(8) FIG. 4B shows inner and outer views of the adjustable eyewear restraint of FIG. 1B in a shortened state in accordance with one example;

- (9) FIG. 5A shows an inner view and a side view of an adjustable eyewear restraint in accordance with one example;
- (10) FIG. 5B shows an inner view and a side view of an adjustable eyewear restraint in accordance with one example;
- (11) FIG. 6A is a perspective view of an adjustable eyewear restraint in accordance with one example;
- (12) FIG. 6B is a perspective view of the adjustable eyewear restraint of FIG. 6A in accordance with one example;
- (13) FIG. 6C is a perspective view of an adjustable eyewear restraint in accordance with one example;
- (14) FIG. 7A is a schematic outer view of a strap of an adjustable eyewear restraint in accordance with one example;
- (15) FIG. 7B is a schematic outer view of a loop formed from the strap of FIG. 7A in accordance with one example;
- (16) FIG. 8A is a schematic outer view of a frictional element of a strap of an adjustable eyewear restraint in accordance with one example;
- (17) FIG. 8B is a schematic outer view of an earpiece retained in the frictional element of FIG. 8A in accordance with one example; and
- (18) FIG. 9 shows various views of a retainer device for coupling an eyewear restraint to an earpiece of an eyewear in accordance with one example.
- (19) These drawings are provided to illustrate various invention aspects and are not intended to be limiting of the scope in terms of dimensions, materials, configurations, arrangements or proportions unless otherwise limited by the claims.

DETAILED DESCRIPTION

(20) While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, it should be understood that other embodiments may be realized and that various changes to the invention may be made without departing from the spirit and scope of the present invention. Thus, the following more detailed description of the embodiments of the present invention is not intended to limit the scope of the invention, as claimed, but is presented for purposes of illustration only and not limitation to describe the features and characteristics of the present invention, to set forth the best mode of operation of the invention, and to sufficiently enable one skilled in the art to practice the invention. Accordingly, the scope of the present invention is to be defined solely by the appended claims.

Definitions

(21) In describing and claiming the present invention, the following terminology will be used.

(22) The singular forms “a,” “an,” and “the” provide express support plural referents in the written description unless the context clearly dictates otherwise. Thus, for example, reference to “a material” includes reference to one or more of such materials and reference to “the subject” refers to one or more subjects.

(23) As used herein, the term “substantially” refers to the complete or nearly complete extent or degree of an action, characteristic, property, state, structure, item, or result. For example, an object that is “substantially” enclosed would mean that the object is either completely enclosed or nearly completely enclosed. The exact allowable degree of deviation from absolute completeness may in some cases depend on the specific context. However, generally speaking the nearness of completion will be so as to have the same overall result as if absolute and total completion were obtained. The use of “substantially” is equally applicable when used in a negative connotation to refer to the complete or near complete lack of an action, characteristic, property, state, structure, item, or result. For example, a composition that is “substantially free of” particles would either completely lack particles, or so nearly completely lack particles that the effect would be the same as if it completely lacked particles. In other words, a composition that is “substantially free of” an

ingredient or element may still actually contain such item as long as there is no measurable effect thereof.

(24) The term “coupled,” as used herein, is defined as directly or indirectly connected. “Directly coupled” is defined as actual physical contact or attachment between two objects, structures, or items. “Indirectly coupled” means that two objects are connected by at least one intermediate object or structure. Objects, structures, or elements described herein as being “adjacent to” each other may be connected, in physical contact with each other, or in close proximity to each other, as appropriate for the context in which the phrase is used.

(25) The terms “first,” “second,” “third,” “fourth,” and the like in the written description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation or use in sequences other than those illustrated or otherwise described herein. Similarly, if a method is described herein as comprising a series of steps, the order of such steps as presented herein is not necessarily the only order in which such steps may be performed, and certain of the stated steps may possibly be omitted and/or certain other steps not described herein may possibly be added to the method.

(26) The terms “left,” “right,” “front,” “back,” “top,” “bottom,” “over,” “under,” and the like in the written description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(27) As used herein, a plurality of items, structural elements, compositional elements, and/or materials may be presented in a common list for convenience. However, these lists should be construed as though each member of the list is individually identified as a separate and unique member. Thus, no individual member of such list should be construed as a de facto equivalent of any other member of the same list solely based on their presentation in a common group without indications to the contrary.

(28) Numerical data, including dimensions, weight, density, etc., may be presented herein in a range format. It is to be understood that such range format is used merely for convenience and brevity and should be interpreted flexibly to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. For example, a numerical range of about 1 to about 4.5 should be interpreted to include not only the explicitly recited limits of 1 to about 4.5, but also to include individual numerals such as 2, 3, 4, and sub-ranges such as 1 to 3, 2 to 4, etc. The same principle applies to ranges reciting only one numerical value, such as “less than about 4.5,” which should be interpreted to include all of the above-recited values and ranges. Further, such an interpretation should apply regardless of the breadth of the range or the characteristic being described.

(29) Any steps recited in any method or process claims may be executed in any order and are not limited to the order presented in the claims. Means-plus-function or step-plus-function limitations will only be employed where for a specific claim limitation all of the following conditions are present in that limitation: a) “means for” or “step for” is expressly recited; and b) a corresponding function is expressly recited. The structure, material or acts that support the means-plus function are expressly recited in the description herein. Accordingly, the scope of the invention should be determined solely by the appended claims and their legal equivalents, rather than by the descriptions and examples given herein.

(30) In this disclosure, “comprises,” “comprising,” “containing” and “having” and the like can have the meaning ascribed to them in U.S. Patent law and can mean “includes,” “including,” and the

like, and are generally interpreted to be open ended terms. The terms “consisting of” or “consists of” are closed terms, and include only the components, structures, steps, or the like specifically listed in conjunction with such terms, as well as that which is in accordance with U.S. Patent law. “Consisting essentially of” or “consists essentially of” have the meaning generally ascribed to them by U.S. Patent law. In particular, such terms are generally closed terms, with the exception of allowing inclusion of additional items, materials, components, steps, or elements, that do not materially affect the basic and novel characteristics or function of the item(s) used in connection therewith. For example, trace elements present in a composition, but not affecting the compositions nature or characteristics would be permissible if present under the “consisting essentially of” language, even though not expressly recited in a list of items following such terminology. When using an open ended term in the written description, like “comprising” or “including,” it is understood that direct support should be afforded also to “consisting essentially of” language as well as “consisting of” language as if stated explicitly and vice versa.

(31) Adjustable Eyewear Restraints

(32) An initial overview of technology embodiments is provided below and specific technology embodiments are then described in further detail. This initial summary is intended to aid readers in understanding the technology more quickly, but is not intended to identify key or essential features of the technology, nor is it intended to limit the scope of the claimed subject matter.

(33) The foregoing detailed description describes the invention with reference to specific exemplary embodiments. However, it will be appreciated that various modifications and changes can be made without departing from the scope of the present invention as set forth in the appended claims. The detailed description and accompanying drawings are to be regarded as merely illustrative, rather than as restrictive, and all such modifications or changes, if any, are intended to fall within the scope of the present invention as described and set forth herein.

(34) FIG. 1A shows an adjustable eyewear restraint **100** coupled to a pair of eyewear **102** in accordance with one example, and FIG. 1B shows an adjustable eyewear restraint **104** coupled to a pair of eyewear **106** in accordance with one example.

(35) Regarding FIG. 1A, the adjustable eyewear restraint **100** comprises a first strap **108** comprising a first adjustment end **110** and a first eyewear coupling end **112**. The first eyewear-coupling end **112** can be coupled to a first eyewear frame portion **114**, such as an earpiece **115** (or the eyewear-coupling end may be directly coupled to another portion of an eyewear frame, such as ski or scuba goggles, for instance). The adjustable eyewear restraint **100** comprises a second strap **116** comprising a second adjustment end **118** and a second eyewear-coupling end **120**. The second eyewear-coupling end **120** can be coupled to a second eyewear frame portion **122**, such as an earpiece **123** of the eyewear **102**. As further described below, the first strap **108** and the second strap **116** are slidably engaged (or engageable) to one another to allow adjustment of a length of the restraint **100** by moving at least one of the first adjustment end **110** and the second adjustment end **120** relative to the other one (see e.g., FIGS. 3A-4B).

(36) Similarly (but somewhat inversely), FIG. 1B shows the adjustable eyewear restraint **104** comprising a first strap **150** comprising a first adjustment end **152** and a first eyewear-coupling end **154**. The first eyewear-coupling end **154** can be coupled to a first eyewear frame portion **156**, such as an earpiece **158**. The adjustable eyewear restraint **104** comprises a second strap **160** comprising a second adjustment end **162** and a second eyewear-coupling end **164**. The second eyewear coupling end **164** can be coupled to a second eyewear frame portion **166**, such as an earpiece **168** of the eyewear **106**. As further described herein, the first strap **150** and the second strap **160** are slidably engaged (or engageable) to one another to allow adjustment of a length of the restraint **104** by moving at least one of the first adjustment end **152** and the second adjustment end **162** relative to the other one (see e.g., FIGS. 3A-4B).

(37) FIG. 1B will not be discussed in as much detail as the configuration of FIG. 1A, but as can be appreciated from comparing the views of FIGS. 1A and 1B (and the following description), the

restraint **104** of FIG. **1B** has an inverse configuration to that of restraint **100** of FIG. **1A**. Meaning that the straps are reversed, but effectively operate the same, and are similarly shaped and sized. Of further note, FIG. **1B** is illustrated to show straps that are slightly larger and thicker than that of FIG. **1A**, which is to illustrate that the straps of the restraint **104** can be comprise a relatively thicker positively buoyant material (e.g., neoprene) having a selected volume and density sufficient to impart positive buoyancy on the pair of eyewear when the restraint **104** and eyewear **106** are in fresh or salt water. For example, assume the pair of eyewear weigh at least 30 grams, then the restraint can be a selected type of material, an overall volume of material, and a density to impart a collective positive buoyancy to the restraint and the eyewear. Thus, if the restraint and eyewear are dropped into fresh or salt water, they will float for easy retrieval. This can also be true for the restraint **100** of FIG. **1A**.

(38) FIGS. **2A-4B** show various views and configurations of the adjustable eyewear restraint **100** of FIG. **1A**. Specifically, FIG. **2A** shows the second strap **116** and FIG. **2B** shows the first strap **108**. FIG. **2A** shows first side view (e.g. inside view) A, a second side view (e.g. outside view B), a top view C, a bottom view D, an eyewear-coupling end view E, and an adjustment end view F, of the second strap **116**. Likewise, FIG. **2B** shows a first side view (e.g. and inside view) view G, a second side view (e.g. an outside view) H, a top view I, a bottom view J, an eyewear-coupling end view K, and an adjustment end view L, of the first strap **108**.

(39) In one example, the first strap **108** comprises the adjustment end **110** that has a first opening **200** (such as a slot or other aperture) formed through the adjustment end **110**. The first opening **200** can be vertically oriented relative to the horizontal length of the strap **108**, and can have a height $H_{sub.1}$ and a width $W_{sub.1}$. The first strap **108** comprises a middle section **202** defined between the adjustment end **118** and the eyewear coupling end **112**. The middle section **202** has a height $H_{sub.2}$ and a thickness $T_{sub.1}$, and the adjustment end **118** has a height $H_{sub.3}$, which is sized larger than $H_{sub.1}$ and sized to allow a wearer to grab a portion of the adjustment end **118**.

Similarly, the second strap **116** comprises the adjustment end **110** that includes a second opening **204** and a third opening **206**, such as parallel slots, formed through the adjustment end **110**. The second and third openings **204** and **206** each have a height $H_{sub.4}$ and a width $W_{sub.2}$, which may be the same or substantially the same dimensions as $H_{sub.1}$ and $W_{sub.1}$ of the first opening **200**. The second strap **116** comprises a middle section **204** between the adjustment end **110** and the eyewear-coupling end **120**. The middle section **204** has a height $H_{sub.5}$ and a thickness $T_{sub.2}$, and the adjustment end **110** has a height $H_{sub.6}$.

(40) The openings **200**, **204**, and/or **206** can be rectangular slots formed vertically as shown, or they can be straight openings or arced/non-linear openings, and can be slits, ellipses, or any other suitable configuration that can receive a middle section of either strap, as the case may be. In one embodiment, the openings do not have a closed perimeter, but rather can include an opening that reaches an edge of the strap to form a “U-shaped open slot” for example. Such an opening at the edge of the strap allows engagement of the other strap by sliding the strap through the open edge. This may be useful in cases where the adjustment ends are semi-rigid or rigid to provide sufficient structural support for an open slot to receive a middle section from an upper or lower area of the slot, for example.

(41) As illustrated best on FIGS. **2A** and **2B**, the straps **108** and **116** are substantially planar along the length of the adjustment ends and the middle sections. This helps to overlay the straps on each other and flat against a wearer's head, which further contributes to the frictional force between the straps because surfaces of the middle sections can be slidably biased against each other to increase the frictional force between the straps.

(42) As further illustrated on FIGS. **3A** and **3B**, a portion of the middle section **202** of the first strap **108** extends through the second opening **204** of the second strap **116** about an inner side **300** of the second strap **116**. A portion of the middle section **202** of the first strap **108** further extends through the third opening **206** about an outer side **302** of the second strap **116**, and through toward the inner

side **300** of the first strap **108**. Thus, a transition section **304** of the first strap **108** extends between the second and third openings **204** and **206**. Similarly, the second strap **116** extends through the first opening **200** about an outer side **306** of the first strap **108**. Accordingly, the first strap **108** is in an interwoven configuration with the second strap **116**, as shown, through the second and third openings **204** and **206**. Therefore, the straps **108** and **116** are slidably interwoven to each other about respective adjustment ends to form a substantially planar retainer configuration. This is best shown on FIGS. **1A** and **1B** where the retainers are collectively substantially flat (in an arc) when overlaying each other on a wearer's head. The openings of the adjustment ends need not be disposed through the adjustment ends. Rather, they could be disposed through an area of the middle section(s) of the straps.

(43) Accordingly, the first and second straps **108** and **116** are slidably engaged to each other and movable relative to each other, bi-directional, as illustrated by arrows X on FIG. **3B** when a wearer pulls the adjustment ends **110** and **118** toward each other or away from each other. In this way, the adjustment end **118** can have a pull tab **310** and the adjustment end **110** can have a pull tab **312**. The pull tabs **310** and **312** can be sized and shaped, as shown, such that a wearer can grip each tab with a respective thumb and index finger, for example, when adjusting the restraint **100**. Any shape, size, or configuration that facilitates or eases grasping by a user can be employed. Advantageously, the pull tabs **310** and **312** are each positioned on the same side, specifically on an outer side of the restraint **100** (e.g. outer, or side facing away from a user's head when the restraint is in use), so that the wearer can readily locate and pull the tabs as desired. In an alternative embodiment, the pull tabs can both be positioned on the other side of the straps (i.e. on the inner or side facing toward a user's head when the restraint is in use) however, and can have an additional finger engagement portion than as shown, for example, elongated or enlarged shapes (e.g. a flange) that allow a user to grasp the tabs above, or below the straps. In some embodiments, the tabs can be on opposite sides of the retainer with one tab on a side facing away from a user's head and the other tab on a side facing toward the user's head. Again, enlarged or flanged shapes can be used to aid a user's grasp on the tabs.

(44) The openings **200**, **204** and **206** are sized and shaped, and selected from a particular material, to provide a frictionally adjustable restraint **100** about a wearer's head. More specifically, middle section **202** is frictionally and slidably coupled to the second and third openings **204** and **206**, and middle section **204** is frictionally and slidably coupled to the first opening **200**. The straps **108** and **116** can be comprised of an elastomeric or highly elastomeric, material or fabric (e.g. neoprene) material having a known friction coefficient such that, when arranged as in FIG. **3A**, a certain amount of pulling force (kinetic friction force) is required to overcome the (static) frictional force between the openings **202**, **204** and the middle section **202** of the first strap **208**. Alternatively, the straps can be made from a less elastomeric material (e.g. leather), which can impart frictional forces between the straps similarly as neoprene straps, or can include the use of an additional piece of friction-adding material. Furthermore, friction can be increased for less elastomeric materials by increasing the number of openings or slots through which the straps pass or are interwoven. In some embodiments, each strap can include between 1 and 5 lots or openings. In other embodiments, the number of openings on each strap can be different and thus allow both pull tabs **310** and **312** to be on the same side of the restraint. For example, one strap can have an even number of openings or slots while the other strap has an even number of openings or slots. In alternative embodiments, when the pull tabs **310** and **312** are on opposite sides of the restraint, the number of openings or slots in each strap can be the same, or can be both an even number or both an odd number.

(45) Of note, the angle or direction at which each strap extends through a respective opening contributes to the frictional force (static and kinetic) existing between the straps and the opening(s) through which the strap(s) extends. For instance, as illustrated on FIG. **3A**, the first strap **108** extends through the second opening **204** from a lower left direction such that the second opening

204 tends to pinch or bind the strap **208**, thereby contributing to the friction between the straps when on a wearer's head.

(46) In some examples, the force required to move or slide one of the adjustment ends relative to a strap can be 0.10 pounds, and up to 2.5 pounds (i.e., 0.20 pounds of collective force to move both adjustment ends, for instance). In some instances, such as with prescription glasses weighing only 15-20 grams, for instance, said force can be less than 0.10 pounds. In other instances with larger, heavier eyewear (e.g., 30 grams or more), said force for each adjustment end can be more than 0.10 pounds, and sometimes greater than 2.5 pounds of force to move an adjustment end. Thus, a particular restraint can be “tuned” for a particular application, such as selecting certain materials for straps, having certain sized openings and strap dimensions, etc.

(47) In some examples, the opening **200** can have the height $H_{sub.1}$ that is shorter than a height $H_{sub.5}$ of the middle section **203** of the second strap **116**. Likewise, width $W_{sub.1}$ can be shorter than thickness $T_{sub.1}$ of the first strap **108**. Said another way, the area defined by the opening **200** can be smaller than a cross sectional area of the middle section **203** of the second strap **116**. In this manner, because neoprene is compliant (and flexible), for example, the opening **200** can effectively “squeeze” or compress the second strap **116** to increase or optimize the frictional force between the straps to a desired frictional force (static and kinetic). The same holds true for the second and third openings **204** and **206** that receive the first strap **108**. In other examples, the openings **200**, **204**, and/or **206** can have an area larger than a cross sectional area of respective middle sections extending through said opening(s), which can be advantageous in examples where the straps are comprised of a more rigid material than neoprene, such as woven fabric, leather, and the other similar materials (or in examples where the adjustment ends and/or openings are comprised of a rigid material, such as certain plastics, rubbers, etc.). In any event, the openings can be sized and shaped to impart a desired frictional force between the straps such that an attached pair of eyewear is restrained on a wearer's head, and such that the restraint **100** is only adjusted when desired by the wearer (i.e., by pulling on the adjustment end(s)).

(48) In some examples, a particular neoprene strap (**108** and **116**, for example) can have a thickness (e.g., $T_{sub.1}$ or $T_{sub.2}$) of 0.5 mm up to 5.0 mm. As known, neoprene straps have a rubber core that is laminated with a polyester or nylon fabric. Therefore, such fabric coating (e.g., inner and outer sides of straps **108** and **116**) will have a known and selectable coefficient of friction, and the rubber core (e.g., within an opening) will have a known and selectable coefficient of friction. Such coefficients are factors when determining the size of the openings, the thickness of the straps, etc.

(49) In some examples, a variety of neoprene materials can be used to form the straps **108** and **116** (for example), such as F Foam, J Foam, K Foam, L Foam, and the like, and combinations thereof. In one example of the straps **108** and **116** being comprised of L Foam neoprene, such material has a density of approximately 0.17-0.21 g/cm^{sup.3} and a compression deflection of 0.28-0.38 kg/m^{sup.3}. Accordingly, assuming each strap is 4 mm thick, 23 cm long, 19 mm tall at the middle section, and approximately 32 mm tall at the adjustment end section, an L Foam neoprene material would necessarily provide sufficient positive buoyancy to a pair of eyewear weighing approximately 30 grams or less, for instance. Moreover, an L Foam neoprene material with a nylon fabric coating would have sufficient friction between the straps about respective openings and strap surfaces to hold a pair of eyewear about a wearer's head.

(50) FIG. 4A shows a first side (e.g. an inner side) view I and a second side (e.g. an outer side) view O of the restraint **100** in a loosened position L, and FIG. 4B shows first side (e.g. an inner side) view I and a second side (e.g. an outer side) view O of the restraint **100** in a tightened position T. This means that, on FIG. 4A, the adjustment ends **110** and **118** are positioned adjacent or proximate each other such that a length of the restraint **100** is at or near its longest possible configuration. Conversely, FIG. 4B shows the adjustment ends **110** and **118** farther away from each other than on FIG. 4A such that a length of the restraint **100** is at or near its shortest (e.g., worn tightest) possible configuration.

(51) FIGS. 5A and 5B illustrate an example of removably attaching straps of a restraint. Each Figure shows an inner view I of a strap and a side view S of said strap. Specifically, a first strap **508** can have a similar shape and configuration as the first strap **108** of FIG. 2B. Accordingly, the first strap **508** can have an adjustment end **518** having a first opening **500**, and a middle section **502** (only partially shown). And, a second strap **516** (FIG. 5B) can have a similar shape and configuration as the second strap **116** of FIG. 2A. Accordingly, the second strap **516** can have an adjustment end **510** having second opening **504** and a third opening **506**, and a middle section **503** (only partially shown). Therefore, the first strap **508** can be slidably and frictionally engaged to the first opening **500**, and the second strap **516** can be slidably and frictionally engaged to the second and third openings **504** and **506**, as similarly shown on FIGS. 3A and 3B.

(52) The first strap **508** can comprise a first male attachment **550** and a first plurality of female attachments **552** (e.g., holes through the strap). Likewise, the second strap **516** can comprise a second male attachment **554** and a second plurality of female attachments **556**. The male attachments **550** and **554** can be posts with a flange that extends from an inner surface of respective adjustment ends **510** and **518**. The male attachments **550** and **554** can be sized and shaped to frictionally fit or snap-fit through a selected female attachment of an opposing strap. For example, the first male attachment **550** can be interlocked to (e.g., removably coupled to) one of the second plurality of female attachments **556**, which removably secures the first strap **508** to the second strap **516**. In this way, a flange **558** of a post **560** of the male attachment **550** is disposed through a female attachment **552** on an opposing side of the strap **516** such that the flange **558** retains the adjustment end **518** to the strap **516**. The male attachment **550** can then be removed by pulling on the adjustment end **518**, for instance, and then coupled to a different female attachment **556** for adjusting a length of a restraint as desired. Likewise, the second male attachment **554** can be interlocked to (e.g., removably coupled to) one of the first plurality of female attachments **552**, which removably secures the second strap **516** to the first strap **508**. In some examples, the male and female attachments are comprised of a semi-rigid material, such as certain plastics and polymers, to ensure proper attachment between the male and female attachments.

(53) FIGS. 6A and 6B illustrate an adjustable eyewear restraint system **600** according to one example. The system **600** can comprise a strap **602** having an eyewear-coupling end **604** slidably coupleable to an earpiece **606** of a pair of eyewear **608**, such as described with reference to FIGS. 1A and 1B. The eyewear-coupling end **604** comprises a loop **610** size and shaped to allow a portion (e.g., an end) of the earpiece **606** to pass through the loop **610** (e.g., see loop **714** of FIG. 7B). In one example, a cross sectional area of the loop **610** is the same size or smaller than a cross sectional area of the earpiece **606**, thus allowing the loop to frictionally (e.g. tightly) engage the earpiece **606**. In this embodiment, the elastomeric material can be selected so as to exert a sufficient frictional force to retain the earpiece in the loop without further assistance. In some embodiments, the inner material of the loop can be a material with a coefficient of friction that further aids in retaining the eyepiece within the loop (e.g. channel, tube, etc.) and the length of the loop can further be used to add or reduce the frictional relationship between the loop and the earpiece. In some embodiments, the loop can have a larger than a cross sectional area of the earpiece **606** such that the loop **610** loosely fits around the earpiece **606**. This allows for the loop **610** to easily engage a variety of sizes of earpieces without the requirement of substantial force during the engagement process. It also prevents the loop **610** from stretching out or failing due to exhaustion or fatigue in the material thus causing the friction fit to be reduced or lost. Thus, having the loop sized to be loose around an earpiece provides longevity of the loop **610** and reliability for repeated uses and/or with different sized earpieces.

(54) In embodiments where the loop **610** is loosely fit around the earpiece **606**, a frictional element **612** can be coupled to the earpiece **606** and positioned adjacent the loop **610** proximate an inner side **614** of the loop **610**. The loop and frictional element can work cooperatively as a system for retaining the eyewear. In one example, the frictional element **612** is an O-ring having a perimeter

body section with a cross sectional area larger than a cross sectional area of an opening of the loop **610** (FIGS. **6A** and **6B**). The frictional element **612** can have an aperture **616** sized and shaped sufficient to frictionally engage the earpiece **606** and retain the earpiece **606** within the loop **610**, thereby preventing the loop **604** from sliding off of the earpiece **606**. In one example, the frictional element **612** can comprise a compliant material that is frictionally and slidably coupled to the earpiece **606** for adjustment of the eyewear coupling end **604**. This means that the frictional element **612** can be slidably moved by a wearer bi-directionally along the earpiece **606** (see arrows Z) such that the eyewear coupling end **604** is also slidably moved about the same direction and distance as the frictional element **612** along the earpiece **606**. Such movement of the eyewear coupling end **604** can therefore adjust a length of the strap **602** of a restraint. Therefore, a pair of frictional elements such as **612** can be incorporated with the adjustable eyewear restraint **100** of FIG. **1A**, which therefore provides an additional mechanism to adjust the straps about a wearer's head to a desired position and tightness, for instance. Any material, size, or shape that can provide sufficient friction to hold in place on an earpiece can be used.

(55) Furthermore, dual-systems for adjusting an eyewear restraint are possible when a pair of frictional elements like **612** (or **662** of FIG. **6C**) are incorporated with the adjustable straps of FIGS. **1A-6B**, thereby combining two systems of adjustability. And, when the adjustable straps are pulled, as on FIG. **3B**, the frictional elements of the present disclosure provide a counter-acting force against the pulling force of the wearer in order to retain the eyewear coupling ends to a pair of eyewear.

(56) FIG. **6C** illustrates an adjustable eyewear restraint system **650** according to one example. The system **650** can comprise a strap **652** having an eyewear coupling end **654** slidably coupleable to an earpiece **656** of a pair of eyewear (e.g., **608**), such as described with reference to FIGS. **1A** and **1B**. The eyewear coupling end **654** comprises a loop **660** (e.g., **714** of FIG. **7B**) sized and shaped to allow a portion (e.g., an end) of the earpiece **656** to pass through the loop **660**. In one example, a cross sectional area of the loop **660** is larger than a cross sectional area of the earpiece **656** such that the loop **660** loosely fits around the earpiece **656** (as with FIG. **6A**). Accordingly, a frictional element **662** can be coupled to the earpiece **656** and can be positioned adjacent the loop **660** proximate an inner side **664** of the loop **660**. In one example, the frictional element **662** is a wedge (or other body) having a perimeter body section with a cross sectional area larger than a cross sectional area of an opening of the loop **660** to retain the earpiece **656** within the loop **654**. The frictional element **662** can have an aperture **666** sized and shaped sufficient to frictionally engage the earpiece **656** and retain the earpiece **656** through the loop **660**, thereby preventing the loop **654** from sliding off of the earpiece **656**. The aperture **666** of the frictional element **662** can extend all the way through the frictional element **662** (as in FIG. **6C**), or it can be a bore (not shown) that stops partially through the frictional element **662** to receive an end of the earpiece **656**. The aperture **666** can be sized to receive smaller earpieces (i.e., prescription frames) as can be seen when comparing FIG. **6A** to FIG. **6C**.

(57) The frictional element **662** can comprise a compliant material and can be frictionally and slidably coupled to the earpiece **656** for adjustment of the eyewear coupling end **654**. This means that the frictional element **662** can be slidably moved by a wearer bi-directionally along the earpiece **656** (see arrows Y) such that the eyewear coupling end **654** is also slidably moved in the same direction and distance as the frictional element **662**. Such movement of the eyewear coupling end **654** can adjust a length of the strap **652**. Therefore, a pair of frictional elements such as **662** can be incorporated with the adjustable eyewear restraint **100** of FIG. **1A**, which therefore provides an additional mechanism to adjust the straps about a wearer's head to a desired position and tightness, for instance.

(58) Notably, regarding FIG. **6A-6C**, the eyewear coupling ends **604** and **654** each include an attachment portion **670** about which edges of the straps are attached together (e.g., sewn), as will be further discussed below. As shown, the attachment portion **670** is positioned in an upwardly facing

manner relative to the earpiece and the wearer's standing position. This is advantageous over existing restraint systems because the attachment portion **670** does not contact the wearer's head (e.g., scalp and ear), which reduces or eliminates chaffing or skin irritation as compared to restraints that have different configurations where an attachment area contacts the wearer, as further discussed herein. In some embodiments that employ such eyewear coupling ends, the retainer can be either adjustable as described herein, or non-adjustable, for example, a single piece of material. (59) FIGS. 7A and 7B illustrate an eyewear-coupling end **700** and method of making the eyewear coupling end **700** (such as the eyewear coupling ends shown and discussed in FIGS. 1A-6C). Here, the eyewear-coupling end **700** is formed from a strap **702** of an eyewear restraint. The strap **702** has a first edge **704** and an opposing second edge **706**, which may be substantially parallel to each other. The first edge **704** may extend along a distance of a length of the strap **702**, such as on FIGS. 2A and 2B. The second edge **706** is shorter than the first edge **704**. As shown on FIG. 7A (in a pre-completed configuration) the strap **702** is cut and shaped like a “key” such that the strap **702** has an elongated middle section **708** that has a consistent length and width and that is terminated proximate the location of the second edge **706**. In other words, the second edge **706** can be positioned on a flange portion **710** at an end of the strap **702**. The flange portion **710** is formed downwardly and substantially perpendicular from a length of the middle section **708** of the strap **702** (but it can be formed at other angles and/or curved edges).

(60) Once the strap **702** is formed as shown on FIG. 7A, the flange portion **710** is rotated upwardly toward the first edge **704** (as shown by the arrow M) such that the first edge **704** and the second edge **706** are in-line and adjacent each other, thereby having proximately a common collective edge region. The first and second edges **704** and **706** can then be attached to each other to form an attachment portion **712** (such as **670** of FIGS. 6A-6C). Accordingly, a loop **714** is thereby formed having an outer opening **716** and an inner opening **718**, which may have a substantially uniform or consistent tubular shape that is flexible and compliant, for instance. The outer opening **716** can receive an earpiece (e.g., FIG. 6A), and the inner opening **718** can allow the earpiece to pass through.

(61) Therefore, the attachment portion **712** is configured such that the first edge **704** (e.g., upper edge) and the second edge **706** (e.g., lower edge) are attached together substantially in-line with the first edge **704**, as shown on FIG. 7B. The resulting configuration provides an attachment portion **712** that is upwardly facing relative to an earpiece when on a wearer. The attachment portion **712** secures the first edge **704** to the second edge **706** by at least one of stitches, adhesive, fasteners (e.g., plastic crimps), liquid plastic (e.g., using UV rays to adhere), and combinations thereof. FIG. 7B shows fabric stitches **718** that attach the edges **702** and **706** together. It will be appreciated from the views that only a section of the first edge **702** is attached to the second edge **706**.

(62) As shown, the attachment portion **712** has a predetermined length that is approximately a length of the second edge **706**, which can be between approximately $\frac{1}{8}$ inch and $\frac{3}{4}$ inch. This is advantageous because existing systems can have loops that are 1 inch or longer, which can cause skin irritation or discomfort. The loop **714** can be sized and shaped to frictionally receive and retain an earpiece, or it can be sized and shaped to loosely receive an earpiece.

(63) Although edge **704** is shown substantially linear, it can have a non-linear or flange portion extending upwardly proximate where it would be attached to the second edge **706** (e.g., FIG. 7A could show a somewhat “T” shape configuration).

(64) FIGS. 8A and 8B illustrate an example of an adjustable eyewear restraint system **800**. The adjustable eyewear restraint system **800** can comprise a strap **802** having an eyewear-coupling end **804** with a loop **806** to receive an earpiece **808**, as similarly described herein. A frictional element **810** can be coupled to the strap **802** adjacent the loop **806** (e.g., on an inner side of the loop as shown). Here, the frictional element **810** is a frictional panel having a pair of openings **812** and **814**. The frictional element **810** can be attached to the strap **802** by a pair of opposing stitches **816** and **818**, or it can be attached by other devices, such as adhesive. As illustrated on FIG. 8B, a tip

portion **820** of the earpiece **808** extends through opening **814** and through opening **812** in an interwoven manner such that a middle portion **822** of the frictional element **810** is exposed and assists to retain or pinch the earpiece **808** against the strap **802**. The openings **812** and **814** can be slits or slots (or other apertures) that are sized and shaped to receive an earpiece while imparting friction between the strap **802**, the frictional element **810**, and the earpiece **808** to prevent the earpiece **808** from falling out on its own. The frictional element **810**, therefore, is configured to retain the earpiece **808**, but also allow the earpiece **808** to be inserted and removed with relative ease by a wearer. For instance, 0.5 pounds of force may be used to insert and removed the earpiece **808** about the frictional element **810**. Here, the openings **812** and **814** are vertical slits relative to an elongated length of the strap **802**, but the openings can be formed at different angles, arcs, and other configurations that can impart frictional force between the strap and the earpiece. In one example, slits or other openings can be formed directly through the strap **802** itself to receive and retain an earpiece.

(65) FIG. **9** shows various views of an example frictional element **900** that can be removably coupled to an earpiece and to a loop of a strap to retain the earpiece to the strap. Here, the frictional element **900** comprises an earpiece retainer **902**. Thus, FIG. **9** shows a side view S, a front view F, and a rear view R of the earpiece retainer **902**. The earpiece retainer **902** can comprise a first end **904** and a second end **906**. The first end **904** can be positioned through a loop **907** of a strap **909** (both **907** and **909** are illustrated in dashed lines), such as the loop and strap of FIG. **7B**. The first end **904** can have an opening **908** sized and shaped to removably attach the earpiece retainer **902** to the earpiece. In this way, the earpiece retainer **902** can be comprised of a compliant material, such as certain plastics, rubbers, and polymers, which allows the opening **908** to expand slightly for a friction fit to an earpiece. The second end **906** can be positioned at least partially within the loop **907** and can have an enlarged interfacing portion **910** that frictionally couples the earpiece retainer **902** within the loop **907**. In this manner, the enlarged interfacing portion **910** can be an outwardly tapered portion that has a cross sectional area larger than a cross sectional area of the loop **907** so that the earpiece retainer **902** cannot be removed through the loop to the right side (of the page) toward a pair of eyewear, but that it can only be removed to the left side away from pair of eyewear when removing the earpiece retainer **902** from an earpiece. This configuration retains an earpiece to the strap **909**. The eyewear retainer **902** can have vertical ribs **912** that are frictionally engaged to an inner surface of the loop **907** to assist with retaining the retainer and attached earpiece to the loop and the strap. In practice, the earpiece retainer **902** is first inserted into the loop from an inner or left side of the loop **907** (as in FIG. **9**), and then an earpiece of eyewear can be inserted through a right opening of the loop **907**, and then the earpiece can be frictionally fit into the opening **908** of the eyewear retainer **902**. These steps can be reversed for removal of the earpiece from the retainer **902**.

(66) In one example there is provided a method of adjusting a length of an eyewear restraint, such as shown and discussed regarding FIGS. **1A-9**. The method can comprise pulling a first adjustment end **118** of a first strap **108** away from a second adjustment end **110** of a second strap **116** to shorten a length of the restraint **100** (or **104**) to tightening an eyewear restraint about a wearer's head when coupled to a pair of eyewear (see e.g., FIG. **1**). The first strap can be slidably and frictionally engaged to the second strap, as discussed specifically regarding FIGS. **3A** and **3B**. The method can comprise pulling the first and second adjustment ends towards each other to lengthen a length of the restraint to loosening the restraint about the wearer's head. The method can comprise disposing a first earpiece of the pair of eyewear through a first loop of the first strap such that the first earpiece extends at least partially through the first loop. The method can comprise coupling a first frictional element (e.g., **612**, **662**, **810**, or **902**) to the first earpiece proximate the first loop to retain the first earpiece through the first loop (e.g., see the discussion of FIGS. **6A-9**). The first frictional element can be slidably adjustable along the first earpiece to adjust a length of the eyewear restraint. Likewise, the method can comprise similar operations to couple a second

frictional element to an opposing earpiece of a pair of eyewear.

(67) In one example there is provided a method of making an eyewear restraint, such as the adjustable eyewear restraints discussed regarding FIGS. 1A-7B. The method can comprise forming a first strap **108** comprising a first adjustment end **118** and a first eyewear-coupling end **112**. Said first eyewear coupling end can be coupleable to a first eyewear frame portion (e.g., FIG. 6A). The method can comprise forming a second strap **116** comprising a second adjustment end **110** and a second eyewear-coupling end **120**. Said second eyewear coupling end can be coupleable to a second eyewear frame portion (e.g., FIG. 6A). Thus, said first strap and said second strap are slidably engageable to one another to allow adjustment of a length of the restraint by moving at least one of the first adjustment end and the second adjustment end relative to the other one, such as described in greater detail regarding FIGS. 3A and 3B. The method can comprise forming a loop (e.g., **714** of FIG. 7B) by securing a first strap edge to a second strap edge to form an attachment portion, wherein the attachment portion is substantially in-line with the first strap edge, as further described in greater detail regarding FIGS. 7A and 7B.

(68) The first and second straps discussed herein can be made of the same material or different materials. In one aspect, they can be made of the same material. In another aspect, they can be made of different materials. The straps can be made of a number of suitable materials. In one embodiment the materials can be flexible materials. In another embodiment they can be elastomeric materials. Non-limiting examples of materials that can be used to make the straps can include neoprene, neogreen, lycra, thermocline, silicone rubber, polyesters, polyamides, polypropylenes, cotton, silk, wool, leather, the like, and combinations thereof.

(69) The adjustable eyewear restraints discussed herein can be adapted to engage a variety of eyewear devices. Non-limiting examples of eyewear can include prescription eyewear, non-prescription eyewear, safety or protective eyewear, swimming eyewear, magnification eyewear, electronic-display eyewear, and the like. In some aspects, the adjustable eyewear restraint can be universal or nearly universal and a single device can be capable of engaging and being used with nearly any type, size, or shape of eyewear.

(70) Further, a variety of anchoring mechanisms can be used in the current technology. Non-limiting examples can include friction-fit mechanisms (such as **130a**, **130b**), closeable loops, cinches, tying mechanisms, snaps, buckles, Velcro, magnets, adhesive, and the like. Accordingly, the anchoring mechanisms can be adapted to temporarily or permanently attach the adjustable eyewear restraint to the intended eyewear. In one embodiment, the anchoring mechanism may simply be a tube that is made from the material of the restraint. When flexible elastomeric material is used for the restraint, the attachment ends may be created by simply folding the strap lengthwise and stitching it together to create a tube. The tube can then be engaged with an earpiece of an eyewear by sliding the tube lengthwise over the end of the earpiece.

(71) The apertures can have any suitable size or geometry. In one aspect, the apertures are adapted to engage the interlocking straps with a degree of friction that provides a secure adjustment of the adjustable eyewear restraint. Accordingly, the adjustable eyewear restraint can be adjusted (i.e. tightened or loosened) by applying a sufficient amount of force to overcome the degree of friction between the interconnecting straps and their respective apertures. Further, the first strap can engage the pair of second apertures in the second strap with a first degree of friction and the second strap can engage the first aperture in the first strap with a second degree of friction. At least one of the first degree of friction, the second degree of friction, or the additive friction between these two interconnecting straps via the respective apertures can be sufficient to secure the eyewear in a desired/intended position on a user. The first degree of friction can be the same as, greater than, or less than the second degree of friction. The degree of friction can be adjusted based on the size and geometry of the apertures in the respective straps or bands, the size and geometry of the respective straps or bands, and/or the materials selected for the respective straps or bands.

Claims

1. An adjustable eyewear restraint, comprising: a first strap comprising a first adjustment end and a first eyewear coupling end, said first eyewear coupling end being coupleable to a first eyewear frame portion; and a second strap comprising a second adjustment end and a second eyewear coupling end, said second eyewear coupling end being coupleable to a second eyewear frame portion; said first strap and said second strap being slidably engageable to one another to allow adjustment of a length of the restraint by moving at least one of the first adjustment end and the second adjustment end relative to the other one, wherein the first strap comprises a first male attachment and a first plurality of female attachments, and wherein the second strap comprises a second male attachment and a second plurality of female attachments, wherein one of the first plurality of female attachments is selectively interlockable to the second male attachment, and one of the second plurality of second female attachments is selectively interlockable to the first male attachment.
 2. The adjustable eyewear restraint of claim 1, wherein the first adjustment end of the first strap includes a first opening slidably frictionally engageable to the second strap, and wherein the second adjustment end of the second strap includes a second opening and a third opening, each of the second and third openings slidably frictionally engageable to the first strap.
 3. The adjustable eyewear restraint of claim 2, wherein each opening is sized and shaped to provide sufficient frictional force, between the opening and the respective strap slidably engageable there through, to retain the straps to each other about a wearer's head.
 4. The adjustable eyewear restraint of claim 3, wherein each opening has a cross sectional area smaller than a cross section area of the respective strap slidably engageable through the respective opening.
 5. The adjustable eyewear restraint of claim 1, wherein each adjustment end comprises a pull tab positioned on the same side of the retainer when the straps are slidably engaged to each other.
 6. The adjustable eyewear restraint of claim 1, wherein each strap comprises at least one of neoprene, neogreen, lycra, thermocline, silicone rubber, polyesters, polyamides, polypropylenes, cotton, silk, wool, leather, and combinations thereof.
 7. The adjustable eyewear restraint of claim 1, wherein the straps are slidably engaged to each other such that a length of the retainer is shortened when the adjustment ends are pulled away from each other, and such that a length of the retainer is lengthened when the adjustment ends are pulled toward each other.
 8. The adjustable eyewear restraint of claim 1, wherein the straps are slidably interwoven to each other about respective adjustment ends to form a substantially planar retainer configuration.
 9. The adjustable eyewear restraint of claim 1, wherein each strap comprises a first edge and an opposing second edge attached together, proximate respective eyewear coupling ends, to form a loop that receives a respective earpiece, wherein the edges are attached to form an attachment portion substantially in-line with the first edge.
 10. The adjustable eyewear restraint of claim 1, further comprising: a frictional element coupled to an earpiece at each of the first and second eyewear coupling ends.
 11. The eyewear restraint system of claim 10, wherein the frictional element comprises a compliant material and has an aperture sized and shaped to receive the earpiece such that the frictional element is frictionally and slidably coupleable to the earpiece for adjustment of the eyewear coupling end.
 12. The eyewear restraint system of claim 10, wherein the compliant material comprises a rubber material.
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